Section 7.4 Atthaseth Kandkwuthithumrong 125010006 T03

11. $\frac{dy}{dx} = \frac{e^x}{e^y}$

$e^y\,dy = e^x\,dx$

$\int e^y\,dy = \int e^x\,dx$

$e^y + C_1 = e^x + C_2 \rightarrow e^y = e^x + C$

12. $\frac{dy}{dx} = 3x^2 e^{-y}$

$e^y\,dy = 3x^2\,dx$

$\int e^y\,dy = \int 3x^2\,dx$

$e^y + C_1 = x^3 + C_2 \rightarrow e^y = x^3 + C$

13. $\frac{dy}{dx} = \sqrt{y}\cos^2\sqrt{y}$

$\frac{1}{\sqrt{y}}\sec^2\sqrt{y}\,dy = 1\,dx$

$\int \frac{1}{\sqrt{y}}\sec^2\sqrt{y}\,dy = \int 1\,dx$

$2\tan\sqrt{y} + C_1 = x + C_2 \rightarrow 2\tan\sqrt{y} = x + C$

17. $\frac{dy}{dx} = 2x\sqrt{1-y^2}$

$\frac{1}{\sqrt{1-y^2}}\,dy = 2x\,dx$

$\int \frac{1}{\sqrt{1-y^2}}\,dy = \int 2x\,dx$

$\sin^{-1}(y) + C_1 = x^2 + C_2 \rightarrow \sin^{-1}y = x^2 + C$

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## Section 8.8

4 $u = 4-x \rightarrow du = -dx \quad \therefore \int_0^4 \frac{dx}{\sqrt{4-x}} = \int_4^0 -\frac{du}{\sqrt{u}} = \int_0^4 \frac{1}{\sqrt{u}}\,du = \left[2\sqrt{u}\right]_0^4 = 4$

10. $u = \frac{x}{2} \rightarrow du = \frac{1}{2}dx \quad \therefore \int_{-\infty}^2 \frac{2\,dx}{x^2+4} = \lim_{b\to-\infty}\int_b^2 \frac{2\,dx}{x^2+4} = \lim_{b\to-\infty}\int_{b/2}^1 \frac{\cancel{4}\,du}{\cancel{4}(u^2+1)} = \lim_{b\to-\infty}\left[\arctan u\right]_{b/2}^1 = \frac{\pi}{4} - \lim_{b\to-\infty}\arctan\left(\frac{b}{2}\right) = \frac{\pi}{4} - \left(-\frac{\pi}{2}\right) = \frac{3\pi}{4}$

14. Let $f(x) = \frac{x}{(x^2+4)^{3/2}} \rightarrow f(-x) = \frac{-x}{((-x)^2+4)^{3/2}} = -\frac{x}{(x^2+4)^{3/2}} = -f(x) \quad \therefore \int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{\infty}\frac{x\,dx}{(x^2+4)^{3/2}} = 0$

20. $u = \tan^{-1}x \rightarrow du = \frac{dx}{1+x^2} \quad \therefore \int_0^{\infty} \frac{16\tan^{-1}x}{1+x^2}\,dx = \int_0^{\pi/2} 16u\,du = \left[8u^2\right]_0^{\pi/2} = 2\pi^2$

23. $u = -|x| \rightarrow du = dx$ (For $x<0$) $\therefore \int_{-\infty}^0 e^{-|x|}\,dx = \lim_{b\to-\infty}\int_b^0 e^{-|x|}\,dx = \lim_{b\to-\infty}\int_b^0 e^u\,du = \lim_{b\to-\infty}\left[e^u\right]_b^0 = e^0 - \lim_{b\to-\infty}e^b = 1 - 0 = 1$

27. $u = \frac{s}{2} \rightarrow du = \frac{1}{2}ds \quad \therefore \int_0^2 \frac{ds}{\sqrt{4-s^2}} = \int_0^1 \frac{2\,du}{\sqrt{4-4u^2}} = \int_0^1 \frac{du}{\sqrt{1-u^2}} = \left[\sin^{-1}(u)\right]_0^1 = \frac{\pi}{2} - 0 = \frac{\pi}{2}$

32. $\int_0^2 \frac{dx}{\sqrt{|x-1|}} = \int_0^1 \frac{dx}{\sqrt{1-x}} + \int_1^2 \frac{dx}{\sqrt{x-1}} = \left[-2\sqrt{1-x}\right]_0^1 + \left[2\sqrt{x-1}\right]_1^2 = 2 + 2 = 4$

35. $\int_0^{\pi/2} \tan\theta\,d\theta = \int_0^{\pi/2}\frac{\sin\theta}{\cos\theta}\,d\theta = \int_1^0 \frac{-1}{u}\,du$ $(u = \cos\theta \rightarrow du = -\sin\theta\,d\theta)$ $= \int_0^1 \frac{1}{u}\,du = \left[\ln u\right]_0^1 = 0 - (-\infty) = +\infty \rightarrow$ Diverges

39. $y = \frac{1}{x} \rightarrow dy = -\frac{1}{x^2}dx = -y^2dx$

$\int_0^{\ln 2} x^{-2}e^{-\frac{1}{x}}\,dx = \int_0^{1/\ln 2} y^2 e^{-y}\left(-\frac{1}{y^2}dy\right) = \int_0^{1/\ln 2} -e^{-y}\,dy = \left[e^{-y}\right]_0^{1/\ln 2} = e^{-1/\ln 2} - 1 \rightarrow$ Converges

41. $\because \sqrt{t} + \sin t \geq \sqrt{t} \quad \forall t \in [0,\pi]$

$\therefore \int_0^{\pi} \frac{1}{\sqrt{t}+\sin t}\,dt \leq \int_0^{\pi}\frac{1}{\sqrt{t}}\,dt = \left[2\sqrt{t}\right]_0^{\pi} = 2\sqrt{\pi} \rightarrow$ Converges

52. $\because \sqrt{x^2-1} < \sqrt{x^2} = x \quad \forall x \in [2,\infty) \quad \therefore \frac{1}{\sqrt{x^2-1}} > \frac{1}{x}$

$\therefore \int_2^{\infty}\frac{dx}{\sqrt{x^2-1}} = \lim_{a\to\infty}\int_2^a \frac{dx}{\sqrt{x^2-1}} > \lim_{a\to\infty}\int_2^a \frac{1}{x}\,dx = \lim_{a\to\infty}\left[\ln|x|\right]_2^a = \lim_{a\to\infty}\ln|a| - \ln 2 = \infty \rightarrow$ Diverges

53. $\because \frac{\sqrt{x+1}}{x^2} \leq \frac{2\sqrt{x}}{x^2} \quad \forall x \geq 1 \rightarrow \int_1^{\infty}\frac{\sqrt{x+1}}{x^2}\,dx \leq \int_1^{\infty}\frac{2\sqrt{x}}{x^2}\,dx = \int_1^{\infty}\frac{2}{x^{3/2}}\,dx = 4\left[\frac{-1}{\sqrt{x}}\right]_1^{\infty} = 4(0+1) = 4 \rightarrow$ Converges

56. $\because \frac{1+\sin x}{x^2} \leq \frac{2}{x^2} \quad \forall x \geq \pi \quad \therefore \int_{\pi}^{\infty}\frac{1+\sin x}{x^2}\,dx \leq \int_{\pi}^{\infty}\frac{2}{x^2}\,dx = \lim_{a\to\infty}\int_{\pi}^{a}\frac{2}{x^2}\,dx = \lim_{a\to\infty}\left[-\frac{2}{x}\right]_{\pi}^{a} = \lim_{a\to\infty}\left(-\frac{2}{a}+\frac{2}{\pi}\right) = \frac{2}{\pi} \rightarrow$ Converges

61. $\because e^x - x > \frac{e^x}{2} \quad \forall x \geq 1 \quad \therefore \int_1^{\infty}\frac{dx}{\sqrt{e^x-x}} < \int_1^{\infty}\sqrt{\frac{2}{e^x}}\,dx = \lim_{a\to\infty}\int_1^a \frac{\sqrt{2}}{e^{x/2}}\,dx = \lim_{a\to\infty}\left[-2\sqrt{2}e^{-x/2}\right]_1^a = \lim_{a\to\infty}\left(-2\sqrt{2}e^{-a/2} + 2\sqrt{\frac{2}{e}}\right) = 2\sqrt{\frac{2}{e}}$

$\rightarrow$ Converges

64. $\because f(x) = e^x + e^{-x}$, $f(-x) = e^{-x} + e^x = f(x) \quad \therefore f(x)$ even, $\int_{-\infty}^{\infty}\frac{dx}{e^x+e^{-x}} = 2\int_0^{\infty}\frac{dx}{e^x+e^{-x}}$

$\because e^{-x} \geq 0 \quad \forall x \geq 0 \rightarrow e^x + e^{-x} \geq 2e^x$, $\int_0^{\infty}\frac{dx}{e^x+e^{-x}} \leq \int_0^{\infty}\frac{dx}{2e^x} = \lim_{a\to\infty}\int_0^a \frac{dx}{2e^x} = \lim_{a\to\infty}\left[-\frac{1}{2}e^{-x}\right]_0^a = \lim_{a\to\infty}\left(-\frac{1}{2}e^{-a}+\frac{1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}$

$\rightarrow$ Converges

65. a) $u = \ln x \rightarrow du = \frac{dx}{x} \quad \therefore \int_1^2 \frac{dx}{x(\ln x)^p} = \int_0^{\ln 2}\frac{1}{u^p}\,du = \begin{cases} \left[\frac{u^{-p+1}}{-p+1}\right]_0^{\ln 2}; p \neq 1 \rightarrow \begin{matrix} p>1 : \text{Diverges} \\ p<1 : \text{Converges} \end{matrix} \\ \left[\ln|p|\right]_0^{\ln 2}; p = 1 \rightarrow \text{Diverges} \end{cases}$ $\therefore$ Converges at $p<1$

b) $u = \ln x \rightarrow du = \frac{dx}{x} \quad \therefore \int_2^{\infty}\frac{dx}{x(\ln x)^p} = \int_{\ln 2}^{\infty}\frac{1}{u^p}\,du = \begin{cases} \left[\frac{u^{-p+1}}{-p+1}\right]_{\ln 2}^{\infty}; p \neq 1 \rightarrow \begin{matrix} p<1 : \text{Diverges} \\ p>1 : \text{Converges} \end{matrix} \\ \left[\ln|p|\right]_{\ln 2}^{\infty}; p = 1 \rightarrow \text{Diverges} \end{cases}$ $\therefore$ Converges at $p>1$

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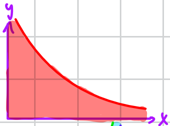
67.




Area $= \int_0^\infty e^{-x}\,dx$

$= \left[-e^{-x}\right]_0^\infty = \lim_{b\to\infty}\left[-e^{-x}\right]_0^b = \lim_{b\to\infty}\left[-e^{-b}+e^0\right] = 0+1 = 1$

72.


a) $V = \int_0^{\pi/2} \pi(\sec^2 x - \tan^2 x)\,dx = \int_0^{\pi/2} \pi \cdot 1\,dx = \pi\left(\frac{\pi}{2}\right) = \frac{\pi^2}{2}$

b) Inner area $= \int_0^{\pi/2} 2\pi \tan x \cdot \sqrt{(dx)^2 + (\sec^2 x\,dx)^2} = \int_0^{\pi/2} 2\pi \tan x \sqrt{1+\sec^4 x}\,dx$

$\geq \int_0^{\pi/2} 2\pi \tan x\,dx = \left[-2\pi \ln|\cos x|\right]_0^{\pi/2} = \infty - 0 = \infty$

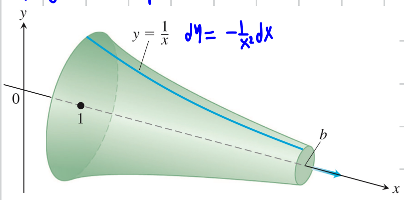
Outer area $= \int_0^{\pi/2} 2\pi \sec x \sqrt{(dx)^2 + (\sec x \tan x\,dx)^2} = \int_0^{\pi/2} 2\pi \sec x \sqrt{1+\sec^2 x \tan^2 x}\,dx$

$\geq \int_0^{\pi/2} 2\pi \sec x\,dx = 2\pi\left[\ln|\sec x + \tan x|\right]_0^{\pi/2}$

$= 2\pi(\infty - 0) = \infty$

Hence, by the comparison test, the area both inner and outer surfaces of the solid have diverged (infinite) value.

76.



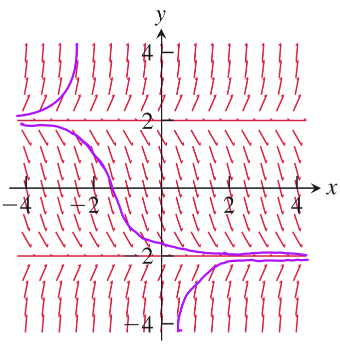

a) $V = \int_1^\infty \pi\left(\frac{1}{x}\right)^2 dx = \pi \int_1^\infty \frac{1}{x^2}\,dx = \pi\left[\frac{-1}{x}\right]_1^\infty = \pi(0+1) = \pi$

b) Area $= \int_1^\infty 2\pi\left(\frac{1}{x}\right)\sqrt{1+\frac{1}{x^4}}\,dx$ which diverges

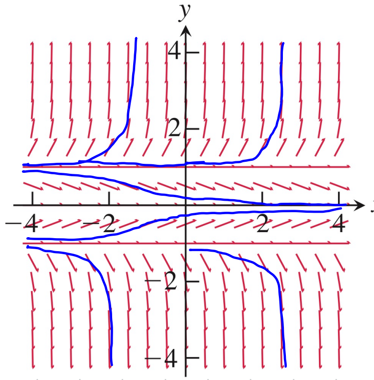
When taking the limit to $\infty$, the modeling no longer remains in the real world, so such contradiction can happen.

Section 9.1

1. (d) (Slope = 0 on line $y=-x$)
2. (c) (Slope = 0 on line $y=-1$, slope same for all $x$ at fixed $y$)
3. (a) (Slope = 0 when $x=0$ and slope undefined (vertical) when $y=0$)
4. (b) (Slope = 0 when $|y|=|x|$)
5.




6.




11. $y' = 1 - \frac{y}{x} = f(x,y)$, $y_0 = -1$, $x_0 = 2$, $dx = 0.5$

$y_1 = y_0 + f(x_0, y_0)\,dx = -1 + \left(1 - \frac{-1}{2}\right)(0.5) = -1 + 0.75 = -0.25$ $\quad (x_1 = 2.5)$

$y_2 = y_1 + f(x_1, y_1)\,dx = -0.25 + \left(1 - \frac{-0.25}{2.5}\right)(0.5) = -0.25 + 0.55 = 0.30$ $\quad (x_2 = 3.0)$

$y_3 = y_2 + f(x_2, y_2)\,dx = 0.30 + \left(1 - \frac{0.30}{3.0}\right)(0.5) = 0.30 + 0.45 = 0.75$ $\quad (x_3 = 3.5)$

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11. (Continue)

$(x>0)$

$\because \frac{dy}{dx} + \left(\frac{1}{x}\right)y = 1 \rightarrow v = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$, $y = \frac{1}{x}\int x \cdot 1\, dx = \frac{1}{x}\cdot\left(\frac{x^2}{2}+C\right) = \frac{x}{2}+\frac{C}{x}$

$\because y(2) = -1 \therefore -1 = \frac{2}{2}+\frac{C}{2} \rightarrow C = -4 \therefore y = \frac{x}{2}-\frac{4}{x}$, $y(3.5) = \frac{7}{4}-\frac{8}{7} = \frac{17}{28} = 0.6071$

14. $y_1 = y_0 + f(x_0,y_0)dx = y_0 + y_0^2(1+2x_0)dx = 1+1^2(1+2(-1))(0.5) = 0.5$, $x_1 = -0.5$

$y_2 = y_1 + f(x_1,y_1)dx = y_1 + y_1^2(1+2x_1)dx = 0.5+(0.5)^2(1+2(-0.5))(0.5) = 0.5$, $x_2 = 0$

$y_3 = y_2 + f(x_2,y_2)dx = y_2 + y_2^2(1+2x_2)dx = 0.5+(0.5)^2(1+2\cdot 0)(0.5) = 0.625$, $x_3 = 0.5$

Actual value: $\frac{dy}{dx} = y^2(1+2x)$

$\frac{1}{y^2}dy = (1+2x)dx \rightarrow -\frac{1}{y}+C_1 = (x+x^2)+C_2$ $y(-1)=1$

$-\frac{1}{y} = x+x^2+C \rightarrow -\frac{1}{1} = -1+(-1)^2+C$, $C = -1 \rightarrow -\frac{1}{y} = (x+x^2)-1$

Actual: $y(0.5) \Rightarrow -\frac{1}{y} = (0.5+0.5^2)-1 = -0.25 \rightarrow y = 4$

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Section 9.2

1. $x\frac{dy}{dx} + 1\cdot y = e^x$

$(xy)' = e^x$

$\int (xy)' = \int e^x\, dx$

$xy = e^x + C \longrightarrow y = \frac{e^x}{x}+\frac{C}{x}$

4. $y' + (\tan x)y = \cos^2 x$

$V = e^{\int \tan x\, dy} = e^{-\ln(\cos x)} = \sec x$ $\left(-\frac{\pi}{2}<x<\frac{\pi}{2} \rightarrow \cos x>0\right)$

$\sec x\cdot y' + \sec x\tan x\cdot y = \cos x$

$\int (y\sec x)' dx = \int \cos x\, dx$

$y\sec x = \sin x + C \longrightarrow y = \sin x\cos x + C\cos x$

16. $t\frac{dy}{dt} + 2y = t^3$ ; $t>0$, $y(2)=1$

$\frac{dy}{dt}+\frac{2}{t}y = t^2 \rightarrow v(t) = e^{\int \frac{2}{t}dt} = e^{2\ln t} = t^2$

$\therefore y = \frac{1}{t^2}\int t^2\cdot t^2 dt = \frac{1}{t^2}\left(\frac{t^5}{5}+C\right) = \frac{t^3}{5}+\frac{C}{t^2}$ $\because y(2)=1 \rightarrow 1 = \frac{2^3}{5}+\frac{C}{2^2} \rightarrow C = -\frac{12}{5}$ $\therefore y = \frac{t^3}{5}-\frac{12}{5t^2}$

17. $\theta\frac{dy}{d\theta} + y = \sin\theta$, $\theta>0$, $y\left(\frac{\pi}{2}\right) = 1$

$\frac{dy}{d\theta}+\frac{1}{\theta}y = \frac{\sin\theta}{\theta} \rightarrow v(\theta) = e^{\int \frac{1}{\theta}d\theta} = e^{\ln\theta} = \theta$

$\therefore y = \frac{1}{\theta}\int \cancel{\theta}\cdot\frac{\sin\theta}{\cancel{\theta}}d\theta = \frac{1}{\theta}(-\cos\theta + C)$ $\because y\left(\frac{\pi}{2}\right) = 1 \rightarrow 1 = \frac{1}{\pi/2}\left(-\cos\frac{\pi}{2}+C\right) \rightarrow C = \frac{2}{\pi}$ $\therefore y = \frac{1}{\theta}\left(-\cos\theta+\frac{2}{\pi}\right)$